Collective Sensemaking in Collaborative GenAI Settings: Exploring How Students Interpret and Negotiate GenAI Outputs in Group Work

Maryam Ayda Sourati

Uppsala University, Department of Informatics and Media, maryam.sourati.7730@student.uu.se

The increasing integration of generative artificial intelligence in education raises important questions about how students interpret, evaluate, and collaboratively work with AI-generated outputs. Drawing on Weick's sensemaking theory, this study explores how university students make sense of generative AI outputs during collaborative group tasks. Using an interpretive qualitative case study approach, 13 students participated in workshops involving text and image generation tasks with ChatGPT and Google Gemini. Data were analyzed using reflexive thematic analysis, resulting in five interconnected themes. Findings show that collaborative sensemaking with GenAI is highly social, shaped by group dynamics and prior AI experience, unevenly distributed, and primarily guided by plausibility rather than factual verification. While GenAI supported ideation and exploration, it simultaneously constrained imagination and shaped what participants considered feasible or worth pursuing. The study contributes to HCI and educational AI research by extending sensemaking theory into collaborative human-AI contexts, highlighting how generative AI systems influence not only task outcomes but also the social and cognitive conditions under which collaborative interpretation occurs.

CCS CONCEPTS • Human-centered computing ~ Collaborative and social computing ~ Empirical studies in collaborative and social computing • Human-centered computing ~ Human computer interaction (HCI) ~ Empirical studies in HCI • Applied computing ~ Education ~ Collaborative learning

Additional Keywords and Phrases: Generative AI, sensemaking, collaborative learning, human-AI interaction, qualitative case study

"This work was submitted in partial fulfilment for a master's degree in Human – Computer Interaction at Uppsala University, Sweden, on 18 May 2026. Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author must be honored.

© 2026 Copyright is held by the owner/author(s)."

1 INTRODUCTION

The use of Generative AI (GenAI) is growing rapidly in collaborative settings such as university group work and collective learning. Despite this growth, GenAI's integration into collaborative workflows remains underdeveloped [32]. There is limited research done on how the incorporation of AI in these collaborative spaces is shaping group interaction and sensemaking practices. As a result, collaborative settings such as universities do not yet have a strategic approach to integrating AI into their workflows.

Existing research indicates both opportunities and challenges related to AI in education and organizations. On one hand, AI has introduced both alternative innovative and personalized learning pathways; on the other hand, it challenges existing evaluation methods [2]. Recent work on generative AI in higher education has primarily focused on individual usage, with little emphasis on how these tools influence collaborative learning or group decision-making [23].

Studies on algorithmic interaction show that people often rely on algorithmic recommendations over human judgment, a phenomenon known as *algorithm appreciation*, which affects how authority is assigned in decision-making processes [16]. This gap highlights the need for a human-centered qualitative investigation of how participants make sense of GenAI.

This thesis investigates how participants interpret and respond to AI-supported collaborative work, with particular attention to how they make sense of outputs and take actions in groups within structured workshop environments. Understanding this matter is especially important given evidence that excessive reliance on AI for cognitive tasks may lead to surface-level processing, reduced critical reflection, and diminished originality [24]. To investigate this issue, the thesis addresses the following research question:

"How do students make sense of generative AI outputs during structured collaborative group tasks in higher education?"

The empirical setting provides a bounded environment in which students engage with generative AI tools (ChatGPT and Google Gemini) during guided collaborative activities. While collaborative AI use certainly occurs in real-world educational and professional contexts, this study focuses on structured task-based interactions to enable detailed observation of sensemaking processes.

As this thesis focuses on contemporary generative systems, it is essential to clarify the terminology. In this thesis, the term "generative AI (GenAI)" refers to systems that can produce new textual or visual outputs and participate in interactive dialogue, such as tools based on large language models. The term "AI" is used more generally when referring to broad topics or discussing earlier work.

2 BACKGROUND

2.1 Generative AI and Human-AI Interaction

Generative AI systems can create original and creative content from existing data, such as images, texts, music, video, code, and other forms of design, unlike earlier AI systems that primarily supported decision-making or classification [21]. Recent scholarship suggests that GenAI is becoming increasingly integral to everyday contexts, and its role in human-AI collaboration is shifting from a tool to a collaborative partner in knowledge work and learning [30].

Despite these advances, the processes underlying GenAI remain largely opaque. Systems such as large language models generate outputs that are difficult for users to understand or explain, creating challenges for how people interpret, evaluate, and interact with AI-generated content [20, 21]. As a result, interaction with GenAI often involves uncertainty, ambiguity, and the need to make sense of outputs without understanding how they were produced.

Thus, while GenAI expands possibilities of human-AI collaboration, its interpretive challenges remain underexplored, particularly in group settings where multiple individuals must jointly interpret and act on GenAI suggestions. Recent reviews of human-centered AI (HCAI) focus mainly on individual human-AI interactions and measurable principles such as explainability, transparency, and trust. However, less attention has been given to collective and networked forms of human-AI sensemaking and to addressing larger questions about human meaning-making and collaborative interpretation in complex environments [5].

2.2 Understanding, Explanation, and Interrogation

In educational contexts, understanding goes beyond knowledge by involving the ability to grasp structure, meaning, and the relationships between concepts, and apply them in new situations [20]. According to McDermott et al., understanding develops through interaction, questioning, and reflection rather than passive information acquisition. Developing such a deeper understanding requires users to "interrogate" explanations and decision-makers rather than passively accept them [20].

McDermott et al. distinguish between questioning and interrogation; questioning is exploratory, but interrogation is a more systematic process that aims to criticize and confront, and to justify and clarify claims [20]. While both are central to understanding, interrogation requires a higher level of active engagement and rejects passive involvement, supporting deeper comprehension.

Interrogation helps reveal hidden assumptions or implicit beliefs and helps surface biases that shape the interpretation of a concept or situation [20]. Effective interrogation of GenAI requires knowledge, a basic understanding of how GenAI works, its limitations, and the skills to critically question the AI outputs and ask appropriate follow-up questions. These skills together help resist blindly offloading to GenAI and promote appropriate trust, carefulness, and willingness to verify AI-generated outputs [20].

This is especially important in collaborative teams since GenAI uses pattern matching and lacks real awareness of context. Human understanding and explanation are context-sensitive and depend on the interaction, questioning, and meaning. Without critical engagement, learners might misinterpret AI outputs or ignore some errors. This can hurt both learning and collaboration. As a result, AI-generated outputs may sound coherent and convincing without being genuinely accurate or explanatory, leading to issues such as hallucinations [20].

At the same time, full transparency in Human-AI interactions is not realistic. Similar to human communication, trust is often built through interpretation and questioning rather than full access to the underlying reasoning or inner thoughts of others [20]. Consequently, meaningful explanations and interpretable outputs become essential for helping users evaluate AI-generated content, identify potential bias, and make informed judgments.

2.3 Human-AI Interaction in Collaborative Settings

A team consists of a group of two or more people working collaboratively on a shared task, where collaborative learning emphasizes interaction, teamwork, and the joint goal pursuit [13, 25]. As GenAI becomes embedded in collaborative environments, it reshapes group dynamics and how teams generate ideas, manage tasks, and make decisions.

Recent research on GenAI in group work mentions the embedded GenAI features in digital collaboration tools such as Mural, Miro, and Microsoft Whiteboard [10]. In group ideation, teams use GenAI in different ways; either early for idea generation or later as a complementary tool. Reported benefits include exposure to alternative perspectives, access to missing knowledge, assistance in structuring and facilitating tasks, and improved idea flow [10].

Looking at educational contexts, AI-powered systems can increase engagement and motivation by recommending personalized materials, adapting tasks to students' skill levels, immediate feedback, and learning support [13]. In professional contexts, GenAI's integration into team workflows has resulted in improved focus, access to relevant knowledge, and increased meeting satisfaction through reduced cognitive overload and administrative burden; in some cases, AI has also acted as a social catalyst, facilitating conversations [25]. Across these contexts, GenAI functions as both a cognitive and organizational support within teams.

At the same time, research also points to challenges related to trust and reliance. Studies show that students may misuse, overuse, or underuse generative AI, highlighting the importance of calibrated trust in human-AI interactions [30]. Similarly, Abisado et al. found that use of AI is shaped by perceived usefulness and accuracy, which in turn influence adoption and reliance. This emphasizes the importance of an HCI perspective to adapt GenAI in education [1].

Beyond tool-based support, AI can act as an active teammate within human groups. Yan et al. introduced AI agents with contrarian and supportive personas into group discussions. Supportive AI enhanced idea development and discussion, whereas contrarian AI reduced psychological safety [33]. The authors describe this phenomenon as a *Social Blindspot*, where AI subtly shapes group norms and emotions through interaction, even if its presence is not recognized.

Despite benefits, AI-supported collaboration introduces important challenges, including uneven participation, interruptions, generic outputs, hallucinations, and dissatisfaction with results [10, 25]. Broader concerns include overreliance, reduced critical thinking, algorithmic bias, and institutional readiness for integration [13]. These issues indicate that successful integration of GenAI requires not only technical improvements but also careful consideration of pedagogical and social factors.

Overall, prior research shows that GenAI actively shapes collaboration by influencing how teams interact, participate, and construct meaning together. However, less is known about how participants interpret AI-generated outputs in real time and how these interpretations shape group action. This highlights a need for further research on collaborative GenAI use through a sensemaking lens.

3 THEORY

3.1 Sensemaking Theory

This thesis adopts sensemaking theory as the primary theoretical lens for examining how participants interpret and respond to GenAI outputs during collaborative tasks. Abduction complements the study by enabling continuous iteration between observations and theoretical concepts.

Sensemaking happens when individuals encounter something unexpected, unbelievable, or surprising, prompting them to "structure the unknown" [17, 31]. These moments may involve "sensebreaking", where existing meaning is disrupted and must be reconstructed [18, p.13]. One key ontological difference in sensemaking is whether sensemaking is viewed as an internal cognitive process or as a socially constructed process [18, p. 6]. In collaborative settings, these dimensions are intertwined, as individual interpretations are shaped, negotiated, and sometimes transformed through group interaction, and meaning becomes collectively stabilized rather than individually formed.

However, it is important to acknowledge that not all unexpected events trigger sensemaking; it occurs when the gap between what people expect and what actually happens is great and significant enough to prompt the individuals or groups to question the situation and consider how to respond [18, p. 14]. This distinction is important in studying GenAI, where not every output leads to reflection, but certain outputs prompt deeper engagement and discussion.

To better understand sensemaking in practice, Weick describes seven characteristics within sensemaking theory that make sensemaking unique and different from other types of explanatory processes: (1) Grounded in identity construction; (2) Retrospective; (3) Enactive of sensible environments; (4) Social; (5) Ongoing; (6) Focused on and by extracted cues; (7) Driven by plausibility rather than accuracy [35, p. 17].

Identity construction refers to how individuals interpret the world through their sense of self, which is continuously shaped through social interaction rather than remaining stable. Sensemaking is both social and retrospective. People negotiate meaning through interaction, typically constructing it after events occur. This process introduces bias and favors plausibility over accuracy [35, pp. 17-30]. Some researchers suggest a prospective aspect, where individuals anticipate future interpretations while acting [18, p. 2].

Another key element is enactment, where individuals not only interpret their environment but also actively shape it through their actions. Enactment challenges the idea of a fixed environment, highlighting that action and situation evolve together. Sensemaking is also ongoing, with no clear starting point; it happens continuously and becomes visible only when individuals pause to interpret events. This interpretation can sometimes enable or constrain action [35, pp. 30-49].

A central mechanism in sensemaking is the extraction of cues. Rather than interpreting everything as a whole, people rely on extracted cues that represent the larger realities. These cues are seen as representative of broader meanings and can include specific wordings, suggestions, or inconsistencies in case of GenAI outputs. Weick explains that an extracted cue is treated as equivalent to the entire data it comes from [35, p. 49].

Thus, sensemaking is driven not by accuracy but by plausibility and coherence, as people look for workable explanations instead of just correct ones. This is particularly relevant in interactions with GenAI, where outputs may appear plausible despite being incorrect or misleading [20]. Moreover, cues can have multiple interpretations, and meaning often arises by linking cues to possible explanations rather than identifying a single correct one [35, pp. 49-62].

3.2 Analytical Lens and Abductive Reasoning

In this study, sensemaking serves as the theoretical backbone through which participants' interpretations of GenAI outputs are examined, particularly in how individuals collectively interpret, question, and integrate AI-generated outputs within collaborative processes. Instead of assuming that complete transparency is possible, this study views explanation as a tool that participants use to evaluate, challenge, or support AI authority. The study, therefore, focuses on how individuals notice and negotiate cues, construct plausible interpretations, and integrate AI-generated responses into ongoing collaborative processes.

Within the sensemaking theory, cue-based sensemaking functions as the central relational construct, aligning with the research question, which focuses on how individuals notice, interpret, and discuss AI-generated outputs within teams. As discussed before, people rarely interpret situations as a whole; instead, they extract cues from a complex matter and use those to construct plausible interpretations [31]. In addition to cues, plausibility acts as a secondary analytical theme, considering that participants evaluate AI-generated outputs to form the most reasonable explanation rather than the most accurate, and how this influences collaborative understanding.

The constructs together offer a lens for coding and analysis of the data but remain open to empirical refinement, allowing additional themes to emerge from the data. Rather than applying theory deductively or generating it inductively, this study follows abductive reasoning, which allows theory and empirical data to be iteratively analyzed and for new insights to emerge through movement between concepts and observed data. In this study, since the outputs focus on text and image generation by the GenAI systems, cues are expected to appear as moments where team members notice,

question, or understand specific parts of outputs during group conversations, such as wording, suggestions, or questions, visual elements, symbols, or unexpected responses.

Abductive analysis allows researchers to form theories by working iteratively on tensions and expectations, or for the existing theories to be reevaluated or extended [29]. The thesis follows the principles introduced by Van Hulst and Visser to guide the abductive research process. These four principles are: (i) embrace surprises, tensions and doubts; (ii) use knowledges creatively to theorize; (iii) craft through methodological bricolage; and (iv) iterate through the research process [29].

Instead of necessarily following strict procedures, the research is open to adapting the methods to the specific needs of the project and the emerging insights. By combining cue-based sensemaking with an abductive analytical approach, this study is able to capture not only what participants do when interacting with GenAI, but also how they construct meaning from these interactions. This provides a coherent framework for analyzing collaborative sensemaking processes and directly supports the investigation of the research question.

4 METHODOLOGY

4.1 Research Strategy

This thesis follows an exploratory qualitative case study approach, focusing on higher education and primarily students. It combines multiple methods and sensemaking theory to investigate participants' sensemaking, their interaction practices, and tensions and insights on agency, authority, and trust in working with GenAI in groups.

A case study is a method in which a deep examination of specific situations is conducted, considering the context and emphasizing qualitative data and analysis [14, p. 156]. Case studies are particularly suitable for phenomena that have unclear boundaries between the phenomenon and the context [23, p. 121], and are an effective method to build a new theory and test or compare existing theories [23, p. 118]. That being said, the case itself needs to be clearly bounded; otherwise, it becomes difficult to maintain focus and coherence in the research.[10, p. 400]. Therefore, the research question is bounded to students' sensemaking within structured collaborative tasks, providing a clearly defined setting.

The study is exploratory in nature, seeking to surface tensions, assumptions, and possibilities, and help understand the novel issue with the hope of informing new design implications in the preliminary research area [14, pp. 159-160]. Abductive research allows researchers to generate strong explanatory possibilities for social phenomena, yet social phenomena remain context-dependent and variable, meaning that "any generalization is a working hypothesis, not a conclusion" [10 p. 254]. Due to this nature, the study avoids generalizing the findings but instead settles for insights into collective sensemaking with GenAI tools, especially considering that individuals may construct different meanings for the same phenomena [8, p. 9].

In line with the abductive research approach described earlier, the case study design enables iterative movement between empirical data and theoretical concepts, allowing surprising or unexpected patterns to inform further interpretation and theoretical refinement.

4.2 Philosophical Positioning

This study is grounded in interpretive epistemology and constructionist ontology. In this view, reality is socially constructed, and knowledge is obtained through human interactions and interpretation rather than discovered as objective truth [8, p. 8, 43]. Interpretivists believe that a comprehensive understanding of social phenomena requires engagement in qualitative fieldwork and specific contexts [29].

Interpretive research is particularly fitting for this study as it focuses on how people make sense of experiences and negotiate meaning within social settings. As Myers notes, meanings are shaped by context, social interaction, and situational factors [23, pp. 67-68, 125]. This approach aligns with sensemaking theory, which views meanings as continuously constructed and negotiated within collaborative practices.




Figures 1, 2: Workshop setting during the collaborative GenAI task sessions

4.3 Research Approach

In this study, abductive reasoning informs the theoretical framework. Unlike deduction and induction, abduction begins with observed consequences or phenomena and works backward to develop the most plausible explanations for unexpected findings. It involves iterative movement between empirical data and theory, allowing researchers to refine interpretations as new insights emerge. As Tavory and Timmermans argue, abduction bridges the divide between purely empirical and purely theoretical research by combining observations with existing knowledge to generate new understandings [31, p. 6].

Abduction complements sensemaking as they both deal with recalibrating expectations when faced with novel, unusual, or unexpected phenomena [32, p. 171]. In this study, data collected from workshops can reveal unexpected patterns leading to conclusions, insights, or design implications. In other words, participants' sensemaking processes create the "surprise", and abductive reasoning helps the researcher iterate and connect observations to theories and suggest explanations that are both reasonable and grounded in reality.

4.4 Data Collection

The data collection was carried out in cooperation with a fellow master's thesis student working under the same supervisor and within the same research theme to facilitate participant recruitment and workshop management. Although the workshops were jointly organized and conducted, each researcher designed their own task activities and followed unique research goals. This thesis focuses only on the tasks and data generated for this specific study, as the other researcher's work focused on interface design in GenAI. Therefore, data collected for the collaborator's study were not used in this thesis, ensuring clear analytical boundaries and alignment with the research question.

Participant ID	Group	Study Program	AI Experience	Used AI in Group Work
P1	G1	Master's in Human-Computer Interaction and User Experience	Very often (daily)	Yes
P2	G1	Master's in Human-Computer Interaction and User Experience	Very often (daily)	Yes
P3	G1	Master's in Architecture and Urban Design	Often (a few times per week)	No
P4	G2	Master's in Business Management (International Business)	I have tried it but don't use it regularly	Yes
P5	G2	Master's in Human-Computer Interaction	Very often (daily)	Yes
P6	G2	Master's in Information Systems	Very often (daily)	Yes
P7	G3	Bachelor's in Dietetics	Often (a few times per week)	Not sure
P8	G3	N/A	Very often (daily)	Yes
P9	G3	Master's in Human-Computer Interaction	Very often (daily)	Yes
P10	G4	Master's in Computer Science	Very often (daily)	Yes
P11	G4	Master's in Civil Engineering (Information Technology)	Very often (daily)	Yes
P12	G4	Master's in Digital Humanities	Very often (daily)	Yes
P13	G4	Master's in Human-Computer Interaction	Very often (daily)	Yes

Table 1: Overview of Participant Demographics and Background

A total of 13 participants were recruited through an online sign-up form and worked in groups of three to four (Table 1). The form collected basic information on participants' prior experience with generative AI and its use in collaborative settings. Four workshop sessions were held, with the first one serving as an online pilot study and the remaining three held in person, using ChatGPT and Google Gemini tools (Figures 1, 2).

The pilot study was used to test the workshop design, task clarity, timing, and whether it is appropriately pitched for the intended purpose and participant pool [15]. As no major changes were required to the task instructions, prompting procedure, or data collection approach, the pilot data were retained and included in the final analysis (discussed further in Section 6.7). The sessions were recorded to capture both screen activity, including GenAI interactions, and audio of group discussions. To accommodate participants' availability, stay under the time limit, and ensure a balanced distribution of sessions between the researchers, some sessions involved participants completing only one task (Appendix D.2).

4.4.1 Task Design

The workshop tasks were designed to elicit sensemaking and abductive reasoning by having the participants interact with GenAI and reflect on AI-generated outputs while completing tasks. Two types of collaborative tasks were designed for the workshops. One task focused on text generation, as previous research shows that most students commonly use GenAI for text-based activities such as content generation, writing, and concept explanation [11, 26].

In addition, an image generation task was included since studies have shown that turning prompts into images creates challenges, as the generative AI models have to interpret the context and intent described in the text prompts [8], potentially resulting in ambiguity and complex moments of sensemaking. Including both tasks allowed the study to capture sensemaking across different forms of AI output.

The tasks were deliberately open-ended, domain-neutral, and ambiguous enough to encourage discussion, interpretation, and negotiation for example terms such as “course policy” and “The Future Classroom with AI”. The ambiguity was crucial for revealing cue-based sensemaking processes, since participants had to interpret AI outputs, consider their meaning, and decide how to proceed.

The collaborative structure of the workshop was inspired by prior work on team-based GenAI use in project settings [23]. In particular, the workshop adopted the team-oriented collaborative mode (Figure 3), in which prompts are discussed and agreed upon by the entire team prior to submission, AI responses are reviewed collectively, and prompts are iteratively revised based on group debate until a final outcome is achieved. This framework encourages continuous shared engagement and makes the process of interpretation and decision-making apparent.

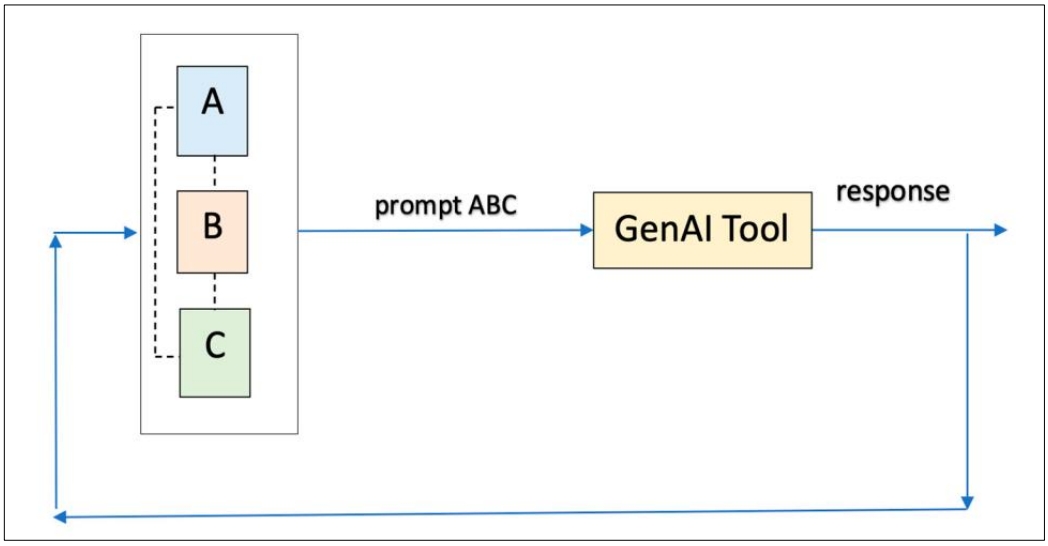


Figure 3: Team-oriented Collaborative Interaction Mode with GenAI: Team members A, B, C interacting with GenAI tool [25].

Task 1: Text Generation

In the first task, participants used ChatGPT to develop a short university guideline containing a set of recommendations for integrating generative AI into course policies. This task required participants to work collaboratively on a single computer and brainstorm using AI, evaluate suggestions, and refine those into three final recommendations as a group. The goal of the task was to prompt participants to engage in conversations and critically interact with the GenAI tool to identify relevant and irrelevant output and to negotiate decisions within their group to reach a final output, thereby engaging in plausibility judgments (Appendix A.1).

Task 2: Image Generation

In the second task, participants used Google Gemini to generate a visual concept for a presentation slide titled "The future classroom with AI". Participants were required to define what the image should represent as a group, evaluate whether the generated image aligned with their expectations, and refine their prompts to finalize the image they would use for such a presentation. This task introduced visual cues and encouraged reflection on how generative models translate abstract concepts into visual form, often leading to ambiguity or unexpected results that required explanation (Appendix A.2).

4.4.2 Interaction Structure and Abductive Reasoning

To support emergence of sensemaking, participants followed a structured interaction approach during both tasks (Appendix A.3). They were asked to think-aloud and discuss their reasoning in groups, to express their expectations of the prompt before generating results (e.g., level of detail, relevance, or format), interpret outputs together, identify unclear or outstanding parts, discuss possible explanations for GenAI behaviors, and test these theories by revising prompts. Each task concluded with participants agreeing on an output they considered complete and sufficiently plausible for use, thereby making an explicit plausibility judgment.

This structure allowed to externalize participants' internal reasoning processes, including cue identification, interpretation, hypothesis generation, and plausibility evaluation, observable in the data, without restricting their approach to the task. This allowed the interaction to remain relatively neutral while still generating relevant data, aligning with qualitative case study perspectives, where cases are not artificial entities or experiments, but rather a part of our reality, from which they cannot be separated, even though they are bounded as cases [10, p. 400].

4.5 Data Analysis

This thesis employed reflexive thematic analysis (RTA), which aligns with the experimental and interpretive approach of this thesis by acknowledging the active role of the researcher in generating themes and interpretations [3]. The analysis followed six steps: (1) data familiarization; (2) coding; (3) initial theme generation; (4) theme development and review; (5) refining, defining and naming themes; and (6) writing up [4, p.43; 5].

The analysis began with familiarization of the dataset through repeated review of recordings, transcripts, and field notes to identify initial observations and codes [4]. In the next stage, the irrelevant sections were filtered out, and a final round of coding was done in relation to the research question to "capture specific and particular meanings within the dataset" [5, p. 86]. Some initial interpretation and concluding notes were also developed at this stage. Themes were then reviewed, refined, and interpreted through an abductive lens, moving back and forth between the data and theoretical concepts to develop the final analytical framework (Figure 4). An example of the coding process is provided in Appendix C.2.

At the next stage, twelve initial themes were developed using an open and iterative interpretive approach. Due to the nature of the reflexive thematic analysis, these themes did not just emerge from the data but were also shaped by the researcher. Through iterative refinement, comparison, merging, and filtering, the initial set of themes was reduced and reorganized in relation to the research question. This resulted in five final themes:

1. Socially and Power-Structured Sensemaking
2. Plausibility-Driven Epistemic Practice
3. Emergent Mental Models and Experience-based AI Theories
4. Prompting as Negotiated Control
5. AI as Cognitive Scaffold and Boundary Shaper



Figure 4: Data transcriptions and Coding Overview

4.6 Ethics

The recruitment form collected limited background information (Table 1), and participation was voluntary, with the right to withdraw and request data deletion. Prior to sessions, all participants provided informed consent for data storage and optional photographs (Appendix D.1). No personal or sensitive information was collected or shared during the workshops. Sessions were recorded locally, capturing both screen activity and discussion audios. Recordings were transcribed using the Whisper local transcription tool, complying with the GDPR regulations, and avoiding transferring data to external services.

The study posed minimal risk, as it involved studying standard educational group work using GenAI tools already commonly encountered in academic settings. Furthermore, the tasks were designed to avoid sensitive or personal topics, and remained low-staked regarding decision-making, and participants were not exposed to deception or manipulation. However, minor risks included potential discomfort for less experienced participants in recorded group settings. This was mitigated by clarifying that the study did not evaluate individual performance but focused on interactional processes.

Motivation to participate was from interest in GenAI, collaborative work, and contributing to research on emerging educational technologies, with informal incentives such as peer interaction and light refreshments. The voluntary participation of the participants might showcase participants' interest or more positive views on GenAI than other users [19]. This may have affected the perspectives represented in the findings.

Looking at the methodological approach, reflexive thematic treats researcher subjectivity as an analytic resource rather than bias, as interpretations are inherently non-neutral [5, p. 45]. Nevertheless, it is necessary to be aware of one's own perspectives. Following Braun and Clarke's emphasis on the importance of reflecting on how researchers "view the research world" [5, p. 53], the researcher's HCI background encouraged a strong focus on interaction, interpretation, user experience, and collaboration rather than the technicalities of the GenAI systems and their performance.

5 RESULTS

Five interconnected themes emerged from the analysis explaining how students understood and interacted with generative AI during collaborative workshop tasks. These themes show how prior experience, group dynamics, and the perceived behavior of AI systems affect the iterative and socially negotiated process of sensemaking.

5.1 Socially and Power-Structured Sensemaking

This theme shows that sensemaking with GenAI is a social process shaped by group interaction, participation asymmetries, and evolving authority structures. Meaning is collectively built but unevenly distributed. Dominant participants, prior knowledge, and communication patterns affect which interpretations become stable, how prompts are created, and which ideas are advanced.

Authority over AI interaction and task direction often developed informally due to reasons such as prior experience or confidence in using GenAI. This affected both prompting strategies and evaluation criteria, as less active members frequently relied on the interpretations of others. In several groups, early contributions had a strong anchoring effect on task direction and collective sensemaking.

For example, in Group 1, P1 immediately delegated the task to GenAI and, by doing so, they shaped the initial interaction with the tool and defined both the approach to the task.

P1: "Honestly, that sounds like a question I would give to ChatGPT."

Similarly, in group 2, P5 introduced and maintained the central idea of the future classroom, and the group followed this vision and continuously built on it.

P5: "I think the formation of the seats is circular…in the middle…this is the AI."

Once introduced, such initial framings are rarely challenged or revoked but rather extended by the group. In one incident, P1 reinterpreted the task as a general policy question. Other members simply affirmed her, which stabilized the understanding around P1, not through equal deliberation.

P1: "…the prompt here does not specify what type of course."

P2: "We can include thesis course."

P1: "I would not. I mean, if the question is, how should a university integrate it into a course policy? I think it's general… so it works for every single course."

Here, meaning is not co-constructed but rather resolved around the dominant participant. In some cases, power dynamics also define or redefine problem definitions. In one case, P1 redirected attention from how humans appear in an AI-generated image introduced by P2, to structural aspects of the image, which the group then adopted.

P2: "They [people in the picture] look very AI."

P3: "Define they look very AI. What do you mean?..."

P1:"I feel like the problem here is not...how the humans look…there's a lack of contrast. There is barely any shadows."

Following this, the group's collective interpretation shifted to P1's diagnosis, illustrating a group's problem definition is not negotiated equally.

P2: "…the room itself looks...Like a 3D model."

P3: "There's no depth to the picture."

However, sensemaking is not always stable. Despite dominant participants, the groups still have moments of negotiation and resistance, such as discussions about a "brain chip," where participants debated consequences and introduced a vote:

P2: "…people can have…brain chip."

P1: "OMG! Would you have it on kids? Do we want to have that future?"

P3: "No."

P1: "Do you know the consequences…cognitive load?...Let's have a demographic vote. Two nos, one yes."

In some cases, a lack of shared understanding led to uncertainty and confusion about the task itself. This highlights that sensemaking is influenced not only by dominance but also by breakdowns in shared understanding, which makes it necessary for groups to rebuild meaning together.

P13: *"Do you want the answer from us or from the AI?"*

P10: "Is it doing homework using AI?...I forgot the aim of the task."

Overall, this theme shows that sensemaking in GenAI-supported group tasks is social, unevenly shared, and shaped by emerging power dynamics. While AI is part of the process, it is the interactions between participants that mainly shape how meaning is created, stabilized, and acted upon.

5.2 Plausibility-Driven Epistemic Practice

Participants primarily assess AI outputs based on plausibility rather than factual correctness, relying on surface cues such as structure, familiarity, and coherence. Despite participants showing concerns about accuracy, bias, and hallucinations, in practice, judgments were guided by coherence, familiarity, and perceived "good enough" quality, while systematic verification was limited and context-dependent. This shows a gap between the stated standards and actual evaluation practices.

Across all sessions, participants rarely assessed GenAI outputs by their accuracy. Instead, they relied on whether the output seemed plausible or aligned with expectations.

P1: "It feels like this is a logical thing to do, like it's common sense… I wouldn't have to fact-check …this seems logical to do."

P3: "Yeah. The rest seem pretty legit honestly."

Participants did not rely on external verification, but instead they judged internally whether outputs "made sense". This was especially clear in Task 1, where outputs were accepted if they matched prior expectations or seemed structurally sound. Group agreement often reinforced this acceptance.

P4: "That was kind of what I envisioned when you started describing this at first."

P5: "Yeah, exactly."

Participants also accepted results despite them being vague or lacking actionable steps:

P12: "It generally looks like a very reasonable [answer]. But there is no specific [instructions]…"

This shows that user satisfaction is not necessarily about output quality, but it is influenced by their desire for efficiency through evaluation. Finalizing task outcome was often influenced by effort, time, and worsening results, rather than full confidence in correctness. This example shows an effort-based satisficing where participants stop rather than optimizing to get the desired outcome.

P1: "I feel like you would still have to black box it in order to fully trust that…But I think this is good enough."

P3: "I think it's a good base."

P1: "Yeah, it's a very good base."

Importantly, participants explicitly acknowledged the limitations of the AI tool, such as hallucinations, bias, unreliable sources, and content overload. However, this skepticism and need for external fact checks remained aspirational rather than practiced.

P1: "Sometimes the page doesn't even exist."

P2: "It makes up links as well."

P1: "I think I would go through all of it and then maybe look at sources myself for each section."

Overall, this theme shows that participants engage in plausibility-driven decision making, judging by intuition, expectations, and efficiency constraints. While participants express a desire for accuracy, their actual practices reflect satisficing strategies that prioritize ease of use over verification.

5.3 Emergent Mental Models and Experience-based AI Theories

This theme captures how participants build changing mental models of GenAI based on their past experiences, interaction patterns, and group support. These models may not necessarily be accurate, but they are practically useful, as they shape expectations, inspire strategies, and influence how users understand the system's behavior and trust outputs.

Across groups, participants relied heavily on experience-based reasoning to predict output characteristics such as quality, structure, and usefulness.

P8: "I think it will be kind of vague. More abstract."

P7: "Like covers most things while not saying anything too specific."

P1: "It's always better the first time…then it gets worse."

A common trend revealed that participants believed prolonged interaction with AI reduces the quality of outputs. As a result, participants changed their interaction strategies by stopping earlier and reducing iterations.

P2: "The more you ask, the more answers will become worse and worse."

P1: "The quality definitely decreases."

Participants also attempt to explain this pattern. Group 1 hypothesized about the reasons behind this behavior, mentioning information overload and focus issues.

P2: "When you give too much context…it becomes worse and worse."

P1: "It's always taking whatever was said before instead of focusing on the most previous thing."

Beyond performance expectations, participants developed broader ideas about how generative AI works. Their views were mainly shaped by their past experiences rather than formal knowledge and generally fell into two types: data-driven models and anthropomorphic tendencies. Some participants described GenAI as pattern-based or constrained by training data.

P1: "I think it's also giving this answer because that's what the data it's trained on is saying. It's probably trained to… always give something about ethics… to make it seem more trustable."

Others personified AI behaviors such as claiming "*it really likes equity*", which showed differing explanations within the groups.

P3: "The AI itself knows that it tends to hallucinate…it has learned it over time."

In other cases, participants formed theories to explain unexpected outputs or to construct a mental model of AI as a strictly literal interpreter of prompts.

P12: "Maybe based on the previous research history."

P10: "Yeah, maybe other universities already have policy about AI tools"

P4: "…it takes things very literally, so it doesn't think outside of the box…it will be literally what we say."

These mental models were not fixed but were revised during interaction, even without verification. Nonetheless, they helped to stabilize understanding. For example, when outputs went against participants' expectations, they used reasoning such as "*maybe it is designed this way*" or "*it runs out of points*" to make sense of it.

Importantly, these models also shaped trust calibration and decision-making. Participants used their theories to decide when to stop prompting, switch chats, doubt outputs, and how much time to invest in editing and verifying. Finally, some participants expressed trust in their teammates because they felt they lacked experience with GenAI tools. For example, P4 expresses a lack of personal experience or disliking AI tools; hence, allocating some decisions to teammates.

P4: "I trust you blindly because I don't use this often…I hate AI."

This theme highlights how mental models, even if individually constructed, are brought into the social dynamics of groups, shaping how participants interact with GenAI rather than reflecting accurate system understanding, and how occasionally replaced by reliance on experienced peers.

5.4 Prompting as Negotiated Control

This theme shows how prompting is not a simple input action but an interaction that allows participants to guide and influence AI behavior. Users gradually build their prompting skills by trying out different words, limits, tones, and formats to shape the responses. This highlights a constant tension between how much control users think they have and the actual unpredictability of the system. Prompting acts as both a technical skill and a way for users to make sense of how AI reacts.

Participants approached prompting as a design-oriented activity requiring precision and refinement.

G1 Prompt 1: "Give the sources you referred to…Split it accordingly into subtitles and give it clear segregation."

G1 Prompt 2: "Add practical guidelines to integrate the recommendations so that it can be implemented better in practice."

G1 Prompt 6: "Do not use any other steps, except the ones stated. Give exact practical steps…"

One participant explicitly reflected on prompting requiring "machine language", while others tried out different wording, tone, and structure. Linguistic choices and meaning became important. For example. P9 claimed the verb "create" is more creative, hence allowing the tool to add things, while "write" is more constrained.

Prompting also served as a way to reduce uncertainty and exercise control. However, control was only partial. Participants often faced misalignment between the prompt and output. This caused frustration and led to iterative adjustments.

P3: "I clearly told it to use points one, two, four, and nine…And it still used seven…"

P1: "Every time we say how…it keeps on just yapping…"

A key aspect of this negotiation was participants' understanding of AI as a literal interpreter. They adapted their prompts accordingly, attempting to reduce ambiguity and unintended interpretations.

P4: "I forgot how literal it interprets everything."

P4: "Maybe we should have said like sphere… because now it interpreted [globe] as the earth."

Control was also exercised beyond wording and in session structure and context management. Participants used new ways to remove unwanted influence from prior outputs, for example, by using a new chat session. The users treated the system as something that accumulates bias over time, and they reset the environment as a solution.

P9: *"I will go to a new chat and start chatting again with a new prompt."*

P7: "…it doesn't have like the luggage from the previous chat. So you can come in with like fresh ideas."

Prompting also turned into a process of collaborative learning. Groups participated in learning through repeated interaction, where each round of prompting helped develop new ideas about how AI would respond. Through this process, participants developed practical strategies such as adding, specifying context, and avoiding ambiguous language.

P2: "Make the characters realistic."

P1: "The image should be utopian."

P2: "Do not include anything else."

These strategies were not pre-existing but emerged through interaction, trial, and negotiation. Overall, prompting functioned as a form of negotiated control under uncertainty, where participants continuously adapted their strategies in response to the system's behavior.

5.5 AI as Cognitive Scaffold and Boundary Shaper

AI served as both a cognitive scaffold that supported idea generation and a constraint on human thinking that shaped the boundaries of what participants considered possible, relevant, or actionable. While GenAI broadened access to concepts and sped up exploration, participants often adjusted their own ideas to align with the system's representation limits.

Across tasks, participants frequently used GenAI as a starting point for ideation, outsourcing the initial thinking process.

P11: *"…first of all, paste everything into ChatGPT and see what happens."*

P13: "I will use GPT to brainstorm…then search myself and decide the structure."

This positioned AI as a cognitive scaffold that shaped how tasks were approached and what counted as a valid starting point. However, this initial scaffolding went beyond as participants increasingly selected from AI-generated suggestions rather than independently defining solution spaces.

P1: "Maybe now we can actually have practical guidelines…I feel like these should be grouped…"

P9: "It gave us eight different perspectives…we didn't have these very clear categories."

P7: "The few ideas I did have were more or less all within this."

Over time, participants also adapted their prompting to previous system limitations, reflecting a shift where participants adjust to GenAI behaviors instead of the tool accompanying them as a neutral tool.

P3: "I think it's better if you kind of limit the amount of prompting you do."

P1: "It's always better the first time, but when you're trying to make changes, that's when it gets worse."

At the same time, participants frequently criticized outputs for being vague or lacking actionable details, positioning GenAI as more useful for idea generation than implementation.

P5: "It's a bit vague for me…I have so many questions."

P12: "It generally looks like a very reasonable [answer]…but there is no specific [instructions]…just some general explanations."

P10: "It's not very accurate… not very specific."

Participants also recognized AI's training data and limitations shaping the outputs and potential bias, leading to repeated or recycled outputs. Despite knowing, the repetition steered participants in a certain direction by creating a boundary, while new possibilities were almost only introduced by participants themselves. Beyond functionality, participants also negotiated the imaginative boundaries of GenAI. Some ideas were abandoned or adjusted based on perceived system limitations, particularly when outputs became difficult to prompt or visualize.

P1: "…maybe even like contact lenses, but I guess we can't visualize it."

P1: "I don't know how to prompt it. Just leave it as that."

P9: "I don't know how to articulate this…"

This resulted in a tendency to favor ideas that aligned with what the system could easily represent, reinforcing a constraint on imagination. Overall, a paradox was evident; GenAI functioned as a boundary-shaping cognitive tool that scaffolded ideation while simultaneously constraining imagination, limiting implementation, narrowing decision spaces, and requiring external validation for completion.

6 DISCUSSION

6.1 Sensemaking as a Social and Uneven Process

While the findings in theme 1 confirm sensemaking as a social act, they challenge the idea of equal collaboration in GenAI contexts. Across groups, meanings were produced through interaction and negotiation; however, contributions depended on participants' prior experience with GenAI, confidence, and prompting skills.

Participants with greater familiarity with GenAI often took leading roles and influenced what was considered relevant, plausible, or actionable. Early contributions frequently became anchors that structured subsequent discussions and interpretations. As a result, collaborative sensemaking became path-dependent, supporting Comes' argument that once decisions and interpretations are enacted, they create path dependencies that narrow future choices [5].

This can be understood through Weick's idea of identity construction [31]. Participants did not view AI outputs neutrally. Instead, their prior confidence and familiarity with technology shaped how they interpreted outputs and how much their interpretations affected others. The findings also support Maitlis and Christianson's view that sensemaking includes forward-looking aspects, where individuals build future expectations while interpreting ongoing events[17].

A key finding is that AI literacy established a form of authority in groups. Participants perceived as more experienced with GenAI often held disproportionate influence, while less experienced members occasionally handed over interpretive responsibility to them. However, this asymmetry extended beyond individual identity.

Authority was reinforced by participants who redirected problem definition and exerted power over what was counted as plausible or worth pursuing. This illustrates how introducing GenAI in education does not automatically grant equal collaboration and learning spaces. Instead, it could reinforce participation inequalities, where experienced users gain greater influence over collective interpretations.

This is highlighted by Comes' claim that AI systems increasingly become participants in the sensemaking dynamics by shaping the cognitive and social foundations on which collective comprehension relies [5]. While Weick highlights social interaction and identity formation [31], this study shows that in AI-supported collaboration, expertise with technology and prompting skills can play major roles in how authority is established and maintained. In this way, the findings extend sensemaking theory into collaborative human-AI settings by highlighting the role of AI literacy in shaping collective interpretation.

At the same time, group interpretations were not always stable. Points of disagreement, confusion, and rethinking showed that sensemaking is an ongoing and adaptable process. Participants sometimes pushed back against dominant views, questioned interpretations, or collectively rebuilt meaning when uncertainty arose. This was especially evident when groups were unsure about their task or doubted whether the AI grasped their intentions. These moments reinforce Weick's view of sensemaking as a continuous process triggered by disruptions, confusion, and breakdowns in expected meanings [31].

6.2 Plausibility Over Accuracy as a Consequence of Opacity

Drawing on theme 2, the findings reveal a critical consequence of GenAI opacity. McDermott et al. emphasize that understanding requires interrogating relationships and structures between concepts [20], yet participants rarely attempted such deep engagement or systematic verifications. Instead, they evaluated whether outputs felt reasonable or aligned with their prior expectations, using phrases such as "*it makes sense*" or "*it seems logical*".

This shows the practical limitations of interrogating systems that are fundamentally not designed to be transparent or encourage questioning. Examples include G1 only confirming the existence of the resources used by the tool, evident in the researcher's field notes: "*They check the resources but are happy only if it exists. They don't check the contents*".

In this sense, GenAI systems generate their own plausibility cues. Unlike Weick's description, which focuses on cues naturally extracted from the environment [31], the data indicated that GenAI tools also created cues, creating a dependency that sensemaking theory does not address. Users extracted meaning not only from the content but also from formatting, fluency, tone, structure, and perceived coherence. These new cues became signals of plausibility, extending Weick's understanding of natural cue extraction to algorithmically generated cues.

While Weick's concept of plausibility over accuracy is proven heavily relevant [31], the findings reveal a critical distinction. In GenAI contexts, plausibility is not just preferred but is a necessary act enforced by constraints revolving around GenAI tools and their opacity. Factors such as time pressure, cognitive overload, group coordination cost, and a tendency to be efficient rather than accurate all contribute to this.

The findings also relate to Logg et al.'s work on algorithm appreciation [16], which shows that people often trust algorithmic recommendations more than human judgment. By producing coherent and persuasive outputs, AI systems shape what participants notice and how they interpret information. Sensemaking becomes partly dependent on signals of plausibility generated by the machine.

The findings also suggest a limitation in Weick's view of plausibility. While plausibility is necessary for action when faced with uncertainty, this study shows that in GenAI interactions, plausibility can become overstated through smooth AI outputs. Participants often accepted outputs, even when they recognized potential inaccuracies, because plausibility helped them make progress on completing tasks.

This study goes further by indicating that reliance on algorithms in collaborative settings is not just about preference or trust. It is socially reinforced through group alignment, efficiency needs, and shared judgments of plausibility. Participants did not blindly trust AI outputs; rather, they were collectively negotiating what level of verification was acceptable given practical constraints.

6.3 Mental Models as Sensemaking Resources

Findings from theme 3 show how participants formed informal theories about how GenAI systems work. They continually built explanations for AI behavior based on past experiences, repeated interactions, and patterns they observed in the outputs. These explanations included beliefs of worsening GenAI outputs over time, interpreting prompts too literally, repeating and recycling ideas, or showing biases embedded within the training data. This aligns with Weick's idea that sensemaking values plausibility more than objective correctness [31].

These mental models were often speculative and unverified, but crucial for stabilizing collaboration and guiding actions. Rather than understanding system mechanics, participants relied on practical, socially useful explanations. This reflects Weick's notion of cue extraction [31], where simple recurring patterns such as "*it gets worse over time*" or "*it's trained on biased data*" stand in as equivalent to a comprehensive understanding.

Assigning human-like behavior to GenAI, such as the AI "*knows that it tends to hallucinate*", "*liking ethics*", or "*getting stuck*", demonstrates how explanations of system behavior were constructed through interpretation rather than technical knowledge. Prompting itself became part of this sensemaking process, and participants increasingly focused on language choices, small wording differences, and formatting choices as influencing outputs.

6.4 Prompting as Recursive Co-Enactment

Theme 4 illustrates how mental models shaped participants' interaction strategies. Beliefs about AI's behavior influenced when groups stopped prompting, restarted conversations, adjusted wording, or dropped ideas altogether. Through Weick's concept of enactment, it is evident that individuals not only interpreted their surroundings but also actively shaped them through their actions [31].

However, in GenAI settings, enactment becomes immediate and recursive. In classical sensemaking theory, environments are shaped through actions and interpreted afterward. In the GenAI context, enactment happened immediately and recursively, as participants moved through rapid cycles of prompting, interpreting outputs, and revising prompts. The environment changed instantly in response to their actions.

Each prompt led to new outputs, interpretations, and revised prompts, which created a recursive chain of meanings. Thus, prompting became a form of recursive co-enactment, where users continuously reshaped the environment that they later interpreted. As Comes argues, AI and humans co-exist within a complex control relationship in which humans steer AI systems while simultaneously adapting their own sensemaking around AI-generated outputs [5].

However, this recursive co-enactment operates in what Manheim calls a "hall of mirrors" [18], where meaning circulates between the users' prompts and the GenAI outputs without being based on the actual system's structure. Participants gradually developed prompting strategies through experimentation rather than formal instruction, learning that certain wording consistently produced different output styles. These insights came from the interactions themselves, indicating that prompting fluency grows as a form of practical knowledge instead of formal expertise.

At the same time, the findings show the limits of this negotiated control. Participants often felt frustrated when prompts didn't produce the expected outcomes. The gap between what they wanted to control and how the system actually behaved highlighted the ongoing lack of clarity in GenAI systems. Even when participants formed theories about the system's behavior, unpredictability remained a significant part of the interaction.

As a result, participants constantly balanced control with awareness of uncertainty, reflecting that control in human-AI systems is not only about those who make decisions, but about how GenAI itself shapes the interpretive conditions in which decisions are made through filtering, structuring, and ordering outputs [5].

6.5 AI as Both a Scaffold and Constraint

Theme 5 reveals a paradoxical role of GenAI in collaborative learning. On one hand, AI served as a cognitive scaffold. Many groups used GenAI as a starting point for brainstorming, which increased access to ideas and facilitated collaboration. This supports earlier studies showing GenAI can help with idea generation and collaboration in educational settings [10, 13].

On the other hand, the same systems limited imagination and shaped what participants thought was possible, relevant, or worth pursuing. More speculative or abstract ideas were often abandoned since participants saw them as hard to prompt or represent using the system. Recent research similarly notes that optimization-driven systems may replace creative ambiguity with predefined structures and patterns, shaping human sensemaking and social norms of interactions [5].

These findings challenge traditional views in sensemaking theory. Weick describes sensemaking as a process where individuals actively build meaningful realities in uncertain situations [31]. However, this study suggests that in collaborative GenAI settings, the limits of what can be understood may be influenced by the features of the system. Participants were not just creating meaning from their environment; they were also changing their thinking to fit the limitations of AI systems. As Comes argues, AI systems not only optimize tasks, but they also actively shape the informational and social environments in which interpretations and decisions occur [5].

Overall, GenAI served as both a scaffold and a boundary-setting tool. Participants often worked within AI-generated categories rather than generating entirely new ones, limiting the range of ideas explored. As a result, collaborative sensemaking became partly limited by the system's representational logic. This also supports worries raised by Riley et al. about reduced critical thinking and less originality in AI-supported cognitive work [24]. Although participants were often aware of the limitations of AI, they adjusted their ideation processes to fit what the system could do.

6.6 Design Implications

The findings suggest that GenAI systems in educational collaboration are not neutral tools but actively shape how groups interpret information, distribute authority, and define what is plausible or worth pursuing. Therefore, design should move beyond usability and efficiency toward supporting critical reflection, equal participation, and meaningful human agency.

A key implication is the need to make uncertainty and limitations more visible. Participants often relied on outputs that felt right without thoroughly verifying them, even when they acknowledged possible hallucinations, bias, or weak reasoning. Future GenAI interfaces should therefore surface uncertainty, missing evidence, or conflicting signals and support interrogability features [20] that promote users to question outputs, compare alternatives, and consider missing information. These features could help reduce superficial acceptance and encourage deeper engagement with AI-generated content.

The findings also stress the importance of supporting collaborative participation rather than individual interaction alone. Since AI literacy often becomes a source of authority in collaborative settings, collaborative GenAI interfaces could, therefore, incorporate shared prompting environments, visible records of prompt changes, or rotating interaction roles to encourage wider participation.

Prompting should also be understood as a collaborative and reflective skill rather than a purely technical one. Educational activities could encourage students to compare prompting strategies, discuss why certain prompts produced particular outputs, and reflect on how wording shaped AI responses.

Finally, GenAI systems should support creativity without prematurely narrowing exploration. Participants often abandoned ideas that were difficult to prompt, visualize, or generate, suggesting that AI can subtly shape what is perceived as possible or worth pursuing. Educational activities and interfaces should therefore encourage revisiting discarded ideas, exploring unconventional alternatives, and reflecting on unexplored possibilities.

Overall, the findings highlight the need for human-centered GenAI design that supports critical reflection, creativity, transparency, and collaborative agency.

6.7 Limitations and Evaluation

Following abductive and interpretive traditions, the study's value lies in the plausibility, relevance, and usefulness of the theoretical insights rather than statistical representativeness [10, p. 253]. A major limitation of the study is its small and limited sample size. While the findings of a single case cannot apply to all students, disciplines, or collaborative contexts, one case study can still be suitable, considering the exploratory qualitative nature of this case study, and the validity of findings does not automatically increase by having multiple cases [23, p. 131].

Another limitation relates to the workshop setting and the inconsistency of the sessions due to participants' availability. While tasks were designed to be collaborative and realistic, they were still structured activities rather than authentic coursework or long-term projects. Real-world educational collaboration may involve higher stakes, longer social relationships, assessment pressures, and evolving group dynamics. The tasks were also designed by the researcher, potentially influencing what participants focused on, questioned, or deemed important.

The study relied on reflexive thematic analysis, which requires interpretation influenced by the researcher's theoretical stance. Furthermore, occasional researcher interventions during workshops encouraged participants to elaborate on their reasoning. This scaffolding approach, while theoretically justified and applied consistently, introduces potential researcher influence. To mitigate this, questions were designed as open-ended as possible and applied uniformly across groups. However, the full extent to which researcher prompting influenced the emergence and framing of themes remains uncertain and represents a limitation. Instead of seeking to eliminate this influence, the study sees reflexivity as an analytical resource, remaining transparent about its effects.

The case study method itself allows the researcher to explore or test theories within the complex real-life situations [23, pp. 128-129], making it particularly appropriate when "how" or "why" questions about contemporary phenomena [7, p. 4]. Hence, it enabled detailed observation of sensemaking as it unfolded during interaction with GenAI systems. However, case studies risk becoming too broad or too detailed [23, pp. 128-131]. To address this, the study relied on iterative refinement of the research questions and careful filtering of data during analysis.

Moreover, qualitative case study research should not be evaluated solely with statistical sampling logic. This study did not intend to produce universal claims, thus, plausibility, richness of evidence, theoretical contribution, and transparency of interpretation is more important than statistical representativeness [23, pp. 131-134].

Another methodological limitation is the lack of multiple data sources for triangulation, which can enhance analytical credibility [14, p. 158]. While workshop recordings, screen interactions, transcripts, and participant discussions offered rich data, the study did not include additional perspectives, such as teachers or institutional stakeholders, or long-term follow-up data.

Despite limitations, the study has several notable strengths. A key strength lies in the richness of qualitative material, which allowed analysis of collaborative sensemaking as it unfolded in real time rather than having participants retrospectively reflect on what happened. Including both text-based and image-based GenAI tasks also enhanced the study by allowing comparisons across two different output forms.

The pilot workshop was included in the final dataset as only minor wording and timing adjustments were made following the pilot, while the overall study design, tasks, and data collection and analysis procedures remained unchanged (Appendices A.1-A.3 and B.1-B.3). Including pilot data in qualitative research can be methodologically appropriate when the pilot session is compatible with the main study design, no substantial procedural changes are introduced, and analytical frameworks remain consistent [12].

Regarding validity and dependability, the study followed an interpretive and abductive research approach in which quality is assessed through the plausibility, relevance, and coherence of the theoretical explanations generated [9]. At the same time, dependability was achieved through clear documentation of workshop procedures, analytical choices, coding rounds, and reflective positioning throughout the research process. Overall, it is crucial to remember that no research approach leads to perfect data collection, and all methods involve trade-offs [14, p. 27].

6.8 Future Work

The findings suggest several directions for future research. First, longitudinal studies could examine how collaborative sensemaking with GenAI changes over time, as the workshops involved relatively brief interactions lasting around 60 to 90 minutes. Research across different fields, workplace teams, teachers, or multidisciplinary groups may uncover different dynamics of authority, negotiation, and AI literacy.

Second, future studies could explore cross-cultural and multilingual sensemaking practices. Since sensemaking is socially and culturally situated [17], norms around authority, collaboration, disagreement, and expertise might affect how different groups engage with GenAI.

Another important direction concerns interface and system design. Future work could experimentally assess design interventions aimed at promoting interrogation, reflection, and collaborative verification practices. This can involve systems with visible uncertainty indicators, team prompting histories, alternative explanations, or structured reflection prompts. Combining workshops with interviews, observations, surveys, or learning outcome measures could further strengthen understanding of collaborative human-AI sensemaking.

Finally, future research should continue to examine the broader educational effects of GenAI. Earlier technologies, such as calculators and grammar correction tools, gradually became part of everyday educational practices. However, generative AI differs significantly as its outputs are probabilistic, opaque, and hard to interrogate when compared to deterministic tools [20]. This raises crucial questions about what cognitive skills might be outsourced over time and how educational systems should balance help, reflection, creativity, and independent reasoning.

7 CONCLUSION

This study reveals that sensemaking in collaborative GenAI settings is a socially negotiated, plausibility-driven process in which participants construct meaning not only from AI-generated outputs but also through interactional dynamics, prompting strategies, and emergent mental models of the system. While Weick's sensemaking framework remains applicable, generative AI introduces new conditions in which cues are algorithmically generated, enactment occurs through prompting, and plausibility becomes a social act to navigate uncertainty and opacity rather than a natural outcome of human interpretation.

USE OF GENERATIVE AI TOOLS IN THE RESEARCH PROCESS

Generative AI and AI-supported tools were used to support language-related tasks, such as helping to polish language. ChatGPT was mainly used to help improve the flow of written sections and brainstorm possible section headings that better reflected the content of each section. Grammarly was used throughout the paper for spelling and grammar corrections during the writing process. AI was also used to transcribe the sessions. However, all transcripts were manually reviewed while re-listening to the recordings in order to correct inaccuracies, unclear wording, or missing sections. The final transcripts used in the analysis were therefore checked and revised by the researcher. In addition, Claude was occasionally used to help identify potential relevant academic sources related to the research topic. However, the included literature was all manually reviewed and selected, and critically used by the researcher. The use of AI was limited to supportive tasks related to writing assistance, organization, transcription support, and literature exploration. AI tools were not used to generate findings, conduct thematic analysis, interpret data, or make analytical decisions. All interpretations, coding decisions, theoretical connections, and final conclusions were developed and evaluated by the researcher.

REFERENCES

[1] Abisado, M. et al. 2025. Trusting the Machine: University Faculty Perspectives on AI in Information Technology Education. *Proceedings of the 26th ACM Annual Conference on Cybersecurity & Information Technology Education* (Sacramento , California , USA, Nov. 2025), 70–75.

[2] Alfredo, R. et al. 2024. Human-centred learning analytics and AI in education: A systematic literature review. *Computers and Education: Artificial Intelligence*. 6, (June 2024), 100215. https://doi.org/10.1016/j.caeai.2024.100215.

[3] Braun, V. and Clarke, V. 2021. Can I use TA? Should I use TA? Should I *not* use TA? Comparing reflexive thematic analysis and other pattern-based qualitative analytic approaches. *Counselling and Psychotherapy Research*. 21, 1 (Mar. 2021), 37–47. https://doi.org/10.1002/capr.12360.
[4] Braun, V. and Clarke, V. 2022. *Thematic analysis: a practical guide*. SAGE.
[5] Comes, T. 2026. Sensemaking AI: Introducing a research and design agenda for human–AI networks. *EPJ Data Science*. 15, 1 (Mar. 2026), 42. https://doi.org/10.1140/epjds/s13688-026-00634-5.
[6] Conaty, F. 2021. Abduction as a Methodological Approach to Case Study Research in Management Accounting — An Illustrative Case. *Accounting, Finance & Governance Review*. 27, (May 2021). https://doi.org/10.52399/001c.22171.
[7] Crotty, M. 1998. *The foundations of social research: Meaning and perspective in the research process*. Routledge.
[8] Elsharif, W. et al. 2025. Visualizing Ambiguity: Analyzing Linguistic Ambiguity Resolution in Text-to-Image Models. *Computers*. 14, 1 (Jan. 2025), 19. https://doi.org/10.3390/computers14010019.
[9] Flick, U. ed. 2022. *The SAGE handbook of qualitative research design*. SAGE.
[10] He, J. et al. 2024. AI and the Future of Collaborative Work: Group Ideation with an LLM in a Virtual Canvas. *Proceedings of the 3rd Annual Meeting of the Symposium on Human-Computer Interaction for Work* (Newcastle upon Tyne United Kingdom, June 2024), 1–14.
[11] Hershkovitz, A. et al. 2025. Task-Centered Analysis of Higher Education Students' Uses of Generative Artificial Intelligence. *Education Sciences*. 15, 12 (Dec. 2025), 1676. https://doi.org/10.3390/educsci15121676.
[12] Kimotho, S. 2025. Piloting Qualitative Research: A Critical Examination of its Role and Practice. *SSRN Electronic Journal*. (2025). https://doi.org/10.2139/ssrn.5271340.
[13] Kovari, A. 2025. A systematic review of AI-powered collaborative learning in higher education: Trends and outcomes from the last decade. *Social Sciences & Humanities Open*. 11, (2025), 101335. https://doi.org/10.1016/j.ssaho.2025.101335.
[14] Lazar, J. et al. 2017. *Research Methods in Human-Computer Interaction*. Elsevier.
[15] Lees, A.B. et al. 2022. Illuminating the Role of Reflexivity Within Qualitative Pilot Studies: Experiences From a Scholarship of Teaching and Learning Project. *International Journal of Qualitative Methods*. 21, (Apr. 2022), 16094069221076933. https://doi.org/10.1177/16094069221076933.
[16] Logg, J.M. et al. 2019. Algorithm appreciation: People prefer algorithmic to human judgment. *Organizational Behavior and Human Decision Processes*. 151, (Mar. 2019), 90–103. https://doi.org/10.1016/j.obhdp.2018.12.005.
[17] Maitlis, S. and Christianson, M. 2014. Sensemaking in Organizations: Taking Stock and Moving Forward. *Academy of Management Annals*. 8, 1 (Jan. 2014), 57–125. https://doi.org/10.5465/19416520.2014.873177.
[18] Manheim, D. 2026. Language Models' Hall of Mirrors Problem: Why AI Alignment Requires Peircean Semiosis. *Philosophy & Technology*. 39, 1 (Mar. 2026), 9. https://doi.org/10.1007/s13347-025-00975-5.
[19] Mastrianni, A. et al. 2025. AI-Enhanced Sensemaking: Exploring the Design of a Generative AI-Based Assistant to Support Genetic Professionals. *ACM Transactions on Interactive Intelligent Systems*. 15, 4 (Dec. 2025), 1–30. https://doi.org/10.1145/3756326.
[20] McDermott, R. et al. 2025. Competency, Understanding and the Role of Explanation in AI-Driven Education. *Learning and Collaboration Technologies* (Cham, 2025), 304–323.
[21] Muller, M. et al. 2024. GenAICHI 2024: Generative AI and HCI at CHI 2024. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Honolulu HI USA, May 2024), 1–7.
[22] Myers, M.D. 2013. *Qualitative research in business & management*. SAGE.
[23] Perifanou, M. and Economides, A.A. 2025. Collaborative Uses of GenAI Tools in Project-Based Learning. *Education Sciences*. 15, 3 (Mar. 2025), 354. https://doi.org/10.3390/educsci15030354.
[24] Riley, C. et al. 2025. Human-AI Interactions: Cognitive, Behavioral, and Emotional Impacts. (2025). https://doi.org/10.48550/ARXIV.2510.17753.
[25] Schätzle, A. et al. 2026. Understanding How Professionals Integrate Generative AI into Team Meetings: A Qualitative Study. (2026).
[26] Sousa, A.E. and Cardoso, P. 2025. Use of Generative AI by Higher Education Students. *Electronics*. 14, 7 (Mar. 2025), 1258. https://doi.org/10.3390/electronics14071258.
[27] Tavory, I. and Timmermans, S. 2014. *Abductive Analysis: Theorizing Qualitative Research*. University of Chicago Press.

[28] Timmermans, S. and Tavory, I. 2012. Theory Construction in Qualitative Research: From Grounded Theory to Abductive Analysis. *Sociological Theory*. 30, 3 (Sept. 2012), 167–186. https://doi.org/10.1177/0735275112457914.
[29] Van Hulst, M. and Visser, E.L. 2025. Abductive analysis in qualitative research. *Public Administration Review*. 85, 2 (Mar. 2025), 567–580. https://doi.org/10.1111/puar.13856.
[30] Wang, F. et al. 2025. In GenAI we trust: An investigation of university students' reliance on and resistance to generative AI in language learning. *International Journal of Educational Technology in Higher Education*. 22, 1 (Aug. 2025), 59. https://doi.org/10.1186/s41239-025-00547-9.
[31] Weick, K.E. 1995. *Sensemaking in organizations*. Sage Publ.
[32] Xiao, Q. et al. 2026. Can GenAI Move from Individual Use to Collaborative Work? Experiences, Challenges, and Opportunities of Coordinating GenAI into Collaborative Newswork. *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems* (Barcelona Spain, Apr. 2026), 1–22.
[33] Yan, L. et al. 2025. The Social Blindspot in Human-AI Collaboration: How Undetected AI Personas Reshape Team Dynamics. arXiv.

A APPENDICES - Main Workshop Protocol

A.1 Task 1: Text Generation with ChatGPT ~25 min

Your team is preparing a short section for a university guideline: "How should a university integrate generative AI into a course policy?"

Use the AI tool to:

- generate ideas
- evaluate suggestions
- refine a short guide.

Your goal: Produce a document including 3 recommendations that you think a university should include in its AI policy for courses.

As a group, decide:

- Which AI suggestions are useful
- Which suggestions should not be included

A.2 Task 2: Image Generation with Gemini ~25 min

Your team needs to create a visual concept for a presentation slide titled: "The Future Classroom with AI."

Use the AI tool to generate an image that represents this idea.

Your goal: Create an image your team would actually use in a presentation.

As a group, decide:

- What the image should show
- Whether the AI result matches your expectations
- How you might improve the prompt

A.3 How to Work During the Tasks

For each task:

1. Before prompting
   Discuss what you expect the AI to produce.
2. Send a prompt to the AI
   Read the output together.

3. Identify anything interesting or surprising
   What stands out? Does it match expectations?
4. Discuss why the AI produced this result
   What might explain the output?
5. Test your idea
   Improve or change the prompt.
6. Decide as a group
   Is the result good enough to use?

Please think aloud and discuss your reasoning out loud while working together.

B APPENDICES - Pilot Session Protocol

B.1 Task 1: Text Generation with ChatGPT (15 minutes)

Your team is preparing a short section for a university guideline: “How should a university integrate generative AI into a course policy?”

Use the AI tool to:

- generate ideas
- evaluate suggestions
- refine a short guide.

Your goal: Produce 3 recommendations that you think a university should include in its AI policy for courses.

As a group, decide:

- Which AI suggestions are useful
- Which suggestions should not be included

B.2 Task 2: Image Generation with Gemini (15 minutes)

Your team needs to create a visual concept for a presentation slide titled: “The Future Classroom with AI.”

Use the AI tool to generate an image that represents this idea.

Your goal: Create an image your team would actually use in a presentation.

As a group, decide:

- What the image should show
- Whether the AI result matches your expectations
- How you might improve the prompt

You may generate more than one image if needed.

B.3 How to Work During the Tasks

For each task:

7. Before prompting
   Discuss what you expect the AI to produce.
8. Send a prompt to the AI
   Read the output together.
9. Identify anything interesting or surprising

What stands out? Does it match expectations?

10. Discuss why the AI produced this result
    What might explain the output?
11. Test your idea
    Improve or change the prompt.
12. Decide as a group
    Is the result good enough to use?

Please think aloud and discuss your reasoning out loud while working together.

C APPENDICES

C.1 Researcher Scaffolding Questions

The following are example guiding questions posed selectively during workshop sessions to externalize and deepen sensemaking and reasoning processes. The questions were designed with the study's theoretical framework in mind and were applied flexibly; groups actively engaged were not interrupted. Questions were framed as open-ended explorations rather than directing participants toward specific answers.

If a group is quiet, ask:
What are you thinking right now?

Trigger deeper sensemaking:

- What do you think the AI misunderstood?
- How could you test that idea?
- What might happen if you changed the prompt?
- Do you think another AI would give the same result?

Trigger cue-based sensemaking:

- What stood out to you in that response?
- Was anything surprising?
- Did the AI respond the way you expected?
- Is there something that doesn't quite make sense?

Trigger abductive reasoning:

- Why do you think the AI produced that answer?
- Do you think this happened because of the prompt or the AI?
- What might explain that output?
- What do you think the AI is assuming here?

Trigger plausibility evaluation

- Would you trust this output if you were using it for real?
- How do you decide the output is final? (trust)

- Is this answer good enough for the task?
- Would you change anything before using it?
- How confident are you in this result?

Image generation task specials

- What part of the image matches your idea?
- What feels off about it?
- Why do you think the AI generated that detail?
- Does this represent learning with AI realistically?
- What would you change?

C.2 Example of Coding and Analysis Process

This is an example illustrating the data transcribed and the coding process (Figure C.2.1), followed by categorizing the codes into themes (Figure C.2.2). The purple notes indicate the initial coding occurring during the transcription controlling process. The pink notes are the final codes generated during the second round of coding, and the yellow notes contain the researcher's initial interpretations of the data, considering the reflexive thematic analysis nature. An anonymized excerpt of the data is also included.

Sample Coded Transcript Excerpt:

[Raw Data]

P7: “Maybe because we didn't really clearly ask for that, but it's not like a clear policy we've gotten from it. It's more like here's a few ideas of how it could be integrated, which of course we can't give them like seven ideas for how they could do one policy. So you would have to continue working on it.”

P9: “I think we kind of, I don't know if it's correct to use the word bias, but we kind of biased it because we said critical thinking and then it started like Socratic tutor, which is critical thinking. And then again, critic assignments, which is critical thinking.”

P7: “We really got stuck on that.”

P9: “Exactly.”

Researcher: “Do you think it's because of the prompt?”

P9: “Yeah.”

P7: “Probably. But also part of it would be how the AI is created, like function and thinking. Because we said a lot of other words that it didn't really get stuck on. It just like the learning better and deeper are vaguer terms, but it kind of skipped that to go full in on critical thinking”

[Coding and Interpretation]

Segment: “...it's not like a clear policy we've gotten from it”

Initial Code: -

Codes:

Expectation mismatch

Output-type misalignment

Task-oriented evaluation

Interpretation:
Expected policy recommendations but got general ideas. This aligns with sensemaking triggered by discrepancy (Weick).
Shift from acceptance → critical assessment.
Breakdowns are not about correctness, but about fitting to task expectations.

Segment: "…we kind of biased it because we said critical thinking and then it started like Socratic tutor…"
Initial Code: Importance of prompting
Codes:
Recognition of prompt-output causality
Reflexive awareness of own influence
Input-output linkage
Interpretation:
Participants realize output is not objective but is shaped by their input. This reflects:
Enactment (Weick): they create the environment they interpret
Abductive reasoning: linking cause (prompt) → effect (output pattern)

Segment: "…part of it would be how the AI is created, like function and thinking. Because we said a lot of other words that it didn't really get stuck on"
Initial Code: Pinpointing the data AI was trained on
Codes:
Selective cue amplification (by AI)
Perceived imbalance in output
Interpretation of AI prioritization
Attribution of "attention" to AI
Interpretation:
Participants interpret AI as choosing certain cues over others. This reflects:
Cue extraction dynamics → not all cues are equal
Construction of AI as an agent with priorities

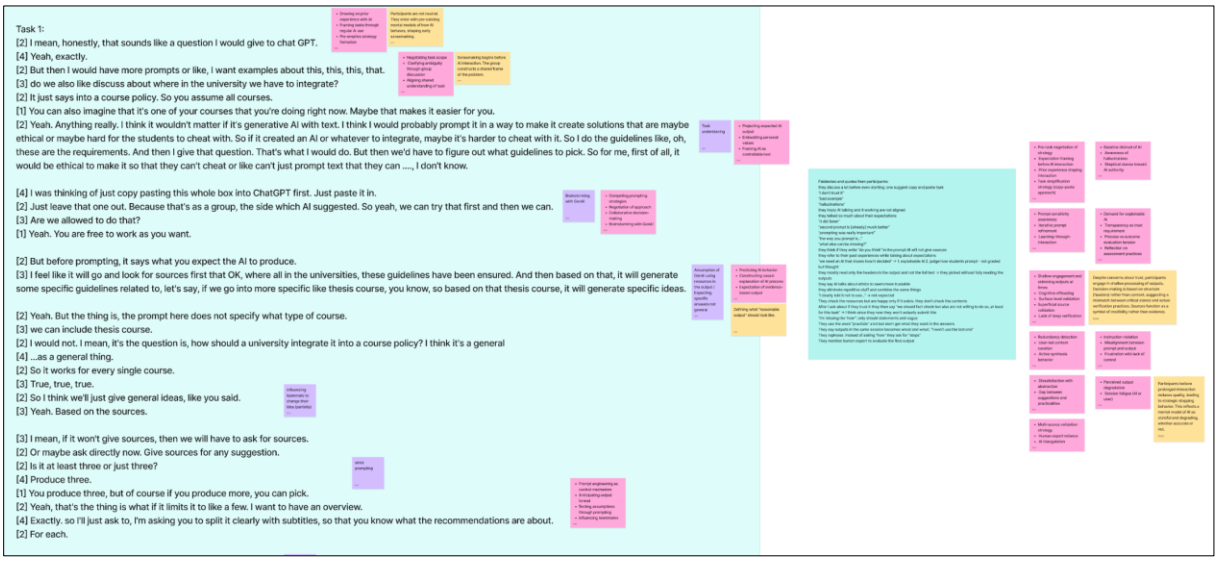


Figure C.2.1: Sample Coding Process

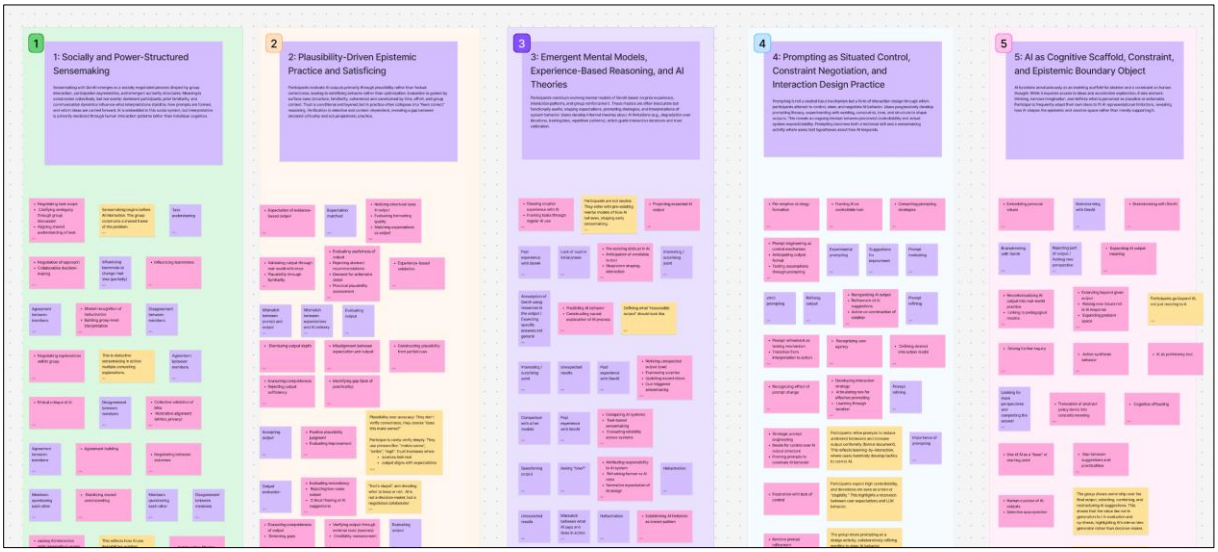


Figure C.2.2: Code Categorization Overview

D APPENDICES

D.1 Consent Form

The online form was required to be signed by participants before joining the session.

Workshop Participation Consent Form

Understanding How Students Make Sense and Interact with GenAI Outputs

Researcher Team: Maryam Ayda Sourati, HCI Master's Student, Uppsala University; Shuai Guo, HCI Master's Student, Uppsala University
Supervisor: Mats Daniels, Department of Information Technology, Uppsala University

Purpose of the Study

This workshop is designed to explore how students interact with and interpret outputs from generative AI tools (ChatGPT, Google Gemini). The goal is to understand the ways students reason, evaluate, and collaborate when using AI, not to assess students' knowledge level of AI.

Data Collection and Storage

During the workshop, we will collect:

- Audio and screen video recordings of your group discussion and interactions
- Prompt history and AI outputs generated during the tasks
- Any written materials you submit during the session
- Photographs of the session (not shared in publications or social media unless you provide consent)

All data will be anonymized for analysis. Your name or other identifying information will not be included in any published material.

Use of Data

The data collected may be used for:

- Transcriptions of the recorded sessions, which may be generated using AI-based transcription services
- Academic research analysis
- Publications or presentations (papers, conferences, reports)
- Using photos in publications or social media such as LinkedIn (Optional)

Data will be stored securely in accordance with GDPR and will only be accessible to the research team and, if required for examination purposes, to the thesis examiner.

Your Rights

- You have the right to withdraw from the study at any time without penalty.
- You may request that your data be deleted at any point.
- You have the right to access your data and request corrections.

Please avoid entering personal or sensitive information into the AI tools during the workshop.

Contact Information

If you have questions about the study or your data, you can contact: maryam.sourati.7730.student.uu.se, shuai.guo.5671@student.uu.se

Consent

Please read each item and tick the box if you agree. To participate in the workshop, you need to agree to items 1 and 2, as these relate to the data necessary for conducting the research study *

☐ I agree to the collection, storage, and use of the data I submit and that is recorded during the workshop for research and academic purposes.

☐ I understand that my data will be anonymized and used in analysis and publications.

☐ (Optional) I consent to photographs (showing me) taken during the workshop being used in academic publications.

Participant Information
Full name: _____
Date: _____

D.2 Data Overview Table

Session	Group	Date	~ Raw Recording Duration	Task(s)
S1	G1	29 March 2026	75 min	1 and 2
S2	G2	30 March 2026	30 min	2
S3	G3	31 March 2026	120 min	1 and 2
S4	G4	2 April 2026	50 min	1
Total: 13 participants, 4 sessions, 275 minutes of recording, ~26900 words of transcript				

Table D.2: Data Collection Overview